Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/28/2025
PI Name:	Santa Maria, Sergio Ph.D.		
Project Title:	ORGANA: Oxidation-Reduction potential and Genetic Assessments for New mission Applications		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sergio.r.santamaria@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-1411
Organization Name:	NASA Ames Research Center		
PI Address 1:	Space Biosciences Research Branch		
PI Address 2:	MS 288-2		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:		**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	01/01/2022	**End Date:**	12/31/2024
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2024 per F. Hernandez/ARC (Ed., 6/21/24). NOTE: End date changed to 09/30/2024 per F. Hernandez/ARC (Ed., 11/1/23).		
Key Personnel Changes/Previous PI:	No changes.		
COI Name (Institution):	Broddrick, Jared Ph.D. (NASA Ames Research Center) Gentry, Diana Ph.D. (NASA Ames Research Center) Liddell, Lauren Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space ionizing radiation (IR) and reduced gravity pose risks to long-duration space travel and eventual non-Earth habitation. Given the difficulties in recreating these effects on Earth, developing an in-depth understanding of these risks before sending crewed missions necessitates fully autonomous biological experiments. As an example, BioSentinel is a small satellite with an integrated BioSensor that will measure metabolic and growth changes induced by deep space radiation in the model organism S. cerevisiae (budding yeast). The Lunar Explorer Instrument for Space Biology Applications (LEIA) project is leveraging the same platform to answer biological questions related to lunar exploration. We hypothesize that both metabolic (redox) and genetic (knockout) assays can distinguish specific changes due to the unique combinations of direct IR damage, indirect oxidative damage, and reduced gravity incurred by exposure to the lunar environment, low Earth orbit (LEO), interplanetary space, and ground-based simulations even before differential survival becomes apparent. These environmental perturbations are known to result in subtle changes in cell growth and activity, but the resulting data is lacking pathway and molecular specificity. By leveraging a metabolic modeling framework and a series of engineered strains, our primary goal is to perform the ground testing necessary to validate strains that yield distinct responses to specific stressors, both laying the groundwork for a potential (and successful) lunar mission to characterize the relative importance of these changes within the capabilities of the BioSensor instrument and improving the usefulness of the alamarBlue assay for past and future missions. To test our hypothesis, we will pursue the following Specific Aims: Aim 1: Derive metabolic pathways usage via modeling of a colorimetric assay. The BioSentinel platform uses a redox dye to detect changes in viability and metabolic activity. AlamarBlue changes color in the presence of cell activity and creates intricate time course data that varies as a function of environmental stress, genetic background, and metabolic activity. To date, the rich information in this data has not been connected explicitly to underlying cellular processes. We will combine the data with metabolic modeling to develop a computational tool that extracts detailed metabolic information. This predictive model will not only enable analysis of our experimental design but can be applied to detect changes caused by lunar-like conditions or any autonomous mission that utilizes this reporter dye. Aim 2: Explore biological effects of a lunar-like environment using DNA repair and stress response defective mutants. Both IR and reduced gravity induce the production of reactive oxygen species (ROS), which play a key role in the generation of cellular and DNA damage, in addition to direct IR damage (e.g., DNA breaks, membrane damage). We will generate yeast mutants defective in DNA repair and stress response pathways, including oxidative damage repair, homologous recombination, and excision repair. Given the short mission duration and the expected low cumulative IR dose, we will also generate mutants containing multiple gene knockouts to increase sensitivity. We will then perform alamarBlue assays after IR exposure in both dry form and in liquid suspension, to study their response. Aim 3: Engineer endogenous ROS scavenging capabilities in yeast. An indirect effect of IR and cellular stress is the buildup of ROS. While S. cerevisiae has several strategies for ROS removal, we will engineer ROS scavenging mechanisms and evaluate if they can mitigate the unique damage caused by spaceflight stress and IR. We will over-express a native ROS scavenging enzyme and engineer a mannitol biosynthetic pathway, which specifically targets an ROS produced by IR. This aim demonstrates a potential mitigation for future spaceflight missions, enabling insight into how engineered scavenging mechanisms could improve crew safety.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ORGANA project (and the investigator team) was the basis for a new mission to the lunar surface, LEIA, which was recently awarded a ~$20M grant as part of NASA's PRISM program. Both projects are trying to understand the effects of the lunar surface environment on biological organisms, both to provide valuable information for future human missions to the Moon and to improve bio production of chemicals of interest like medicine or food. Thus, our project aims to provide lessons on how to improve instrumentation for future missions beyond low Earth orbit (LEO) as well as valuable data for future missions including biology.
Task Progress:	Aim 1: Derive metabolic pathways usage via modeling of a colorimetric assay. We have continued work on wet lab assays to add data into a metabolic model (e.g., glucose uptake, oxygen consumption, redox changes, etc.). In addition, we have completed the design and development of the MVISE device, a 3D-printed, custom sensor-embedded instrument to measure metabolic parameters, including redox state, conductivity, pH, oxygen content, absorbance, etc. Currently, we are performing material biocompatibility and cell growth experiments. Multiple units will be built to perform radiation experiments at Brookhaven National Laboratory next Spring, ~ April-May 2025. Aim 2: Explore biological effects of a lunar-like environment using DNA repair and stress response defective mutants. We have now completed ~2 years of long-term desiccation yeast storage. From this original 40+ yeast strains, we have selected 4 strains that will be used as part f the LEIA mission science verification test (SVT), which will be performed early in 2025. Additional strains are being developed for the final 8 strains that will fly on LEIA. These strains will be tested at the upcoming Brookhaven National Laboratory (BNL) run in the Spring 2025, and as part of the long-term biocompatibility test using flight-like hardware (fluidic cards, optical ground support equipment, etc.). Aim 3: Engineer endogenous reactive oxygen species (ROS) scavenging capabilities in yeast. The first series of strains overexpressing antioxidant enzymes are currently being tested for desiccation tolerance and ionizing (gamma) radiation response (e.g., superoxide dismutases). A second set of strains carrying exogenous countermeasures (i.e., expression of tardigrade genes) should be completed in the upcoming weeks. Like Aim 2, these strains will be tested for long-term desiccation and biocompatibility, and at Brookhaven National Lab. Even though expression of tardigrade genes was not originally proposed in this study, it is one strategy we are using in support of the LEIA lunar mission, and we are already testing them for desiccation and ionizing radiation tolerance.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Articles in Other Journals or Periodicals	Rahmanian S, Slaba TC, Straume T, Bhattacharya S, Santa Maria SR. "Meet BioSentinel: the first biological experiment in deep space." Front. Young Minds. 2024 March 6; 12:1227860. https://doi.org/10.3389/frym.2024.1227860 , Mar-2024

Articles in Peer-reviewed Journals	Lambacher A, Akiyama K, Lingam N, Shuman D, Santa Maria SR. "LEIA: NASA's first biological mission on the lunar surface since 1972." Acta Astronautica. 2025 Feb 27; 231:148.https://doi.org/10.1016/j.actaastro.2025.02.050 , Feb-2025
Articles in Peer-reviewed Journals	Lingam N, Akiyama K, Lambacher A, Shuman D, Gilbert R, Santa Maria SR. "NASA's BioSentinel deep space CubeSat mission: Successes and lessons learned." Acta Astronaut. 2025 Nov;236:188-93. https://doi.org/10.1016/j.actaastro.2025.07.008 , Nov-2025
Articles in Peer-reviewed Journals	Kanapskyte A, Hawkins EM, Liddell LC, Bhardwaj SR, Gentry D, Santa Maria SR. "Space biology research and biosensor technologies: past, present, and future." Biosensors. 2021 Feb;11(2):38. https://doi.org/10.3390/bios11020038 , Feb-2021
Articles in Peer-reviewed Journals	Rahmanian S, Slaba TC, George S, Braby LA, Bhattacharya S, Straume T, Santa Maria SR. "Galactic cosmic ray environment predictions for the NASA BioSentinel Mission, part 2: Post-mission validation." Life Sci Space Res. 2024 October 16. https://doi.org/10.1016/j.lssr.2024.10.006 ; PMID: 39864906 , Oct-2024
Articles in Peer-reviewed Journals	Singh A, Santa Maria SR, Gentry DM, Liddell LC, Lee JA. "Agent-based Model for Microbial Populations Exposed to Radiation (AMMPER) simulates yeast growth for deep-space experiments." Gravit Space Res. 2024 Nov 22;12(1):159-76. https://doi.org/10.2478/gsr-2024-0012 , Nov-2024
Articles in Peer-reviewed Journals	Lapointe MR, Laframboise T, Pirkkanen J, Tai TC, Lees SJ, Santa Maria SR, Tharmalingam S, Boreham DR, Thome C. "Protracted exposure to a sub-background radiation environment negatively impacts the anhydrobiotic recovery of desiccated yeast sentinels." Health Phys. 2024 Jun 1;126(6):397-404. http://dx.doi.org/10.1097/HP.0000000000001804 ; PMID: 38568172 , Jun-2024